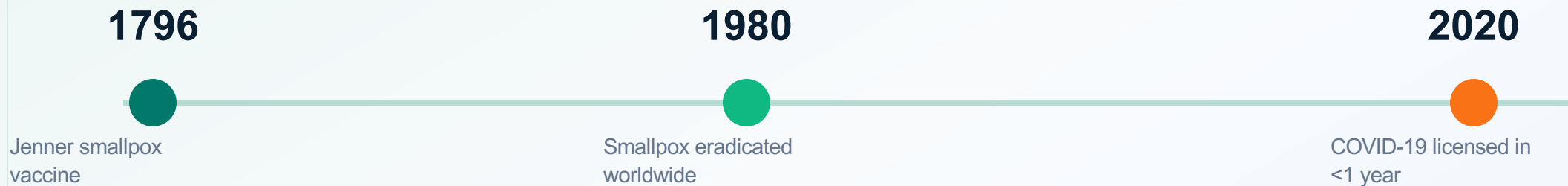


The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines transformed public health by eliminating or drastically reducing deadly diseases over the past two centuries.



Scope: 1796–2020 | Audience: public health professionals and policy analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Gap from pathogen identification to vaccine licensure, selected diseases

Years from identification to licensure



COVID-19 visually emphasized as the record timeline

105 years

Typhoid fever represents the early era of long development timelines.

<1 year

COVID-19 marks the shortest gap in the brief: identified and licensed in 2020.

- Modern tools include advanced microscopy, genome sequencing, mRNA technology, and virus-like particles.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

03

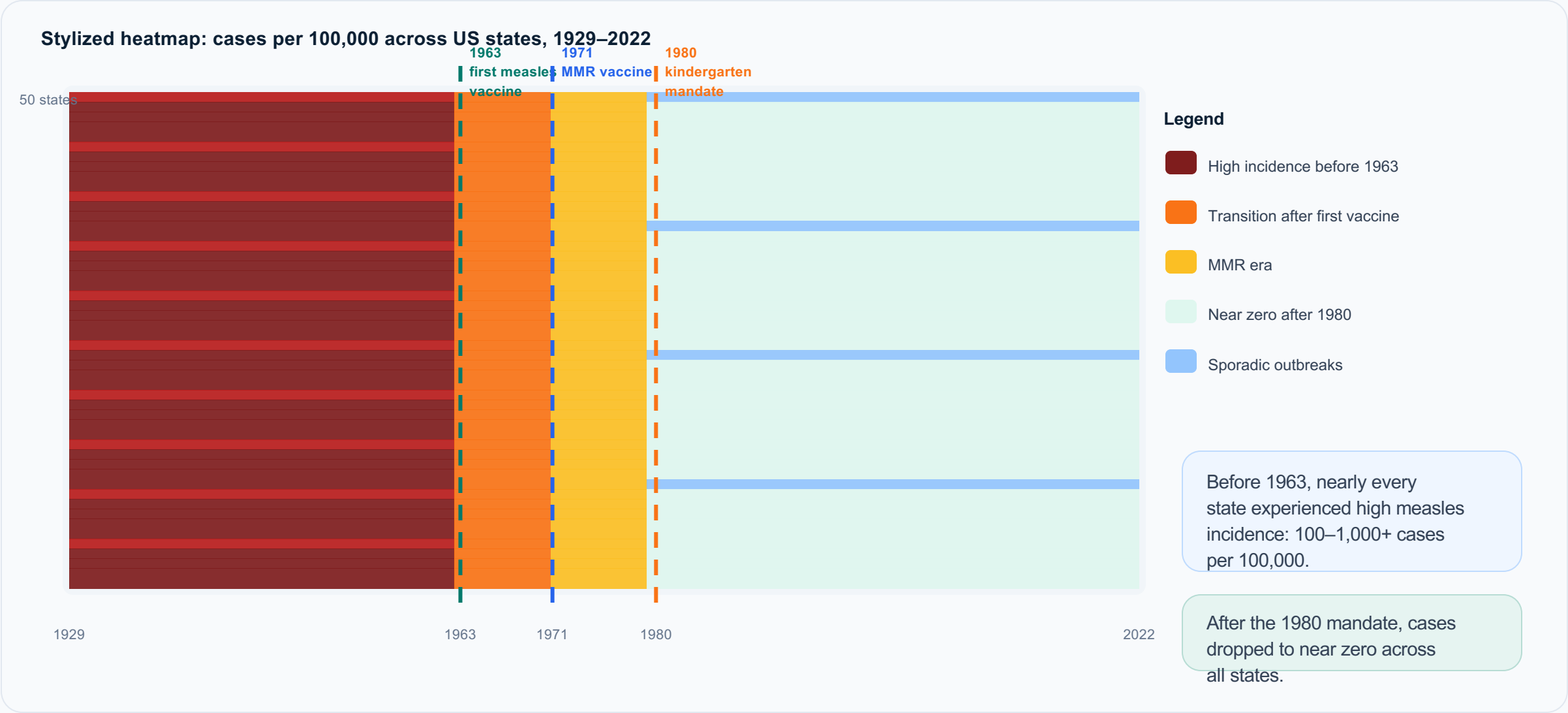
Pre-vaccine vs. post-vaccine burden, per million people per year

Disease	Pre cases	Post cases	Case reduction	Pre deaths	Death reduction	Status
Diphtheria	158	0	100%	13.7	100%	Eliminated
Measles	3,044	0.2	99.99%	2.5	100%	Near-zero cases
Smallpox	250	0	100%	2.9	100%	Eradicated
Acute polio	141	0	100%	10	100%	Eliminated
Pertussis	1,534	52	96.6%	30.8	99.7%	Reduced
Rubella	242	0.04	99.98%	0.09	100%	Near-zero cases
Hepatitis A	465	51	89%	0.5	88.7%	Gap remains
Acute hepatitis B	273	44	83.9%	1	83.6%	Gap remains
Pneumococcal	233	139	40.5%	24	31.3%	Largest gap

Clearly marked: 100% case + death reduction rows are shaded green

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap-style state/time view with milestone annotations



Smallpox: The Only Human Disease Eradicated Through Vaccination

A global campaign moved from resolution to eradication certification in two decades

Eradication timeline



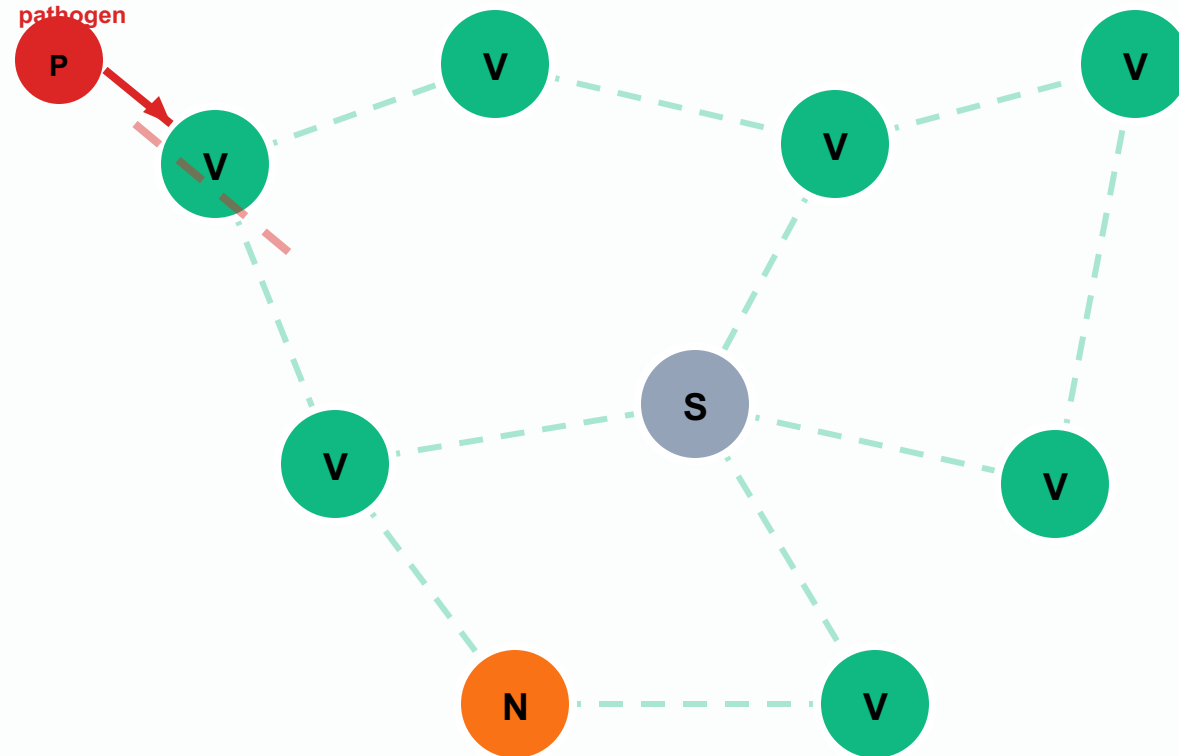
Only human disease
fully eliminated through vaccination

- Why eradication was feasible**
- Virus infected only humans
 - Symptoms were distinguishable
 - Vaccine was cheap and effective

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

06

Population-level vaccination interrupts spread and shields people who cannot be vaccinated



V = vaccinated barrier | N = newborn / vulnerable | S = susceptible

Core mechanism

When enough people are vaccinated, pathogens cannot spread effectively through the population.

Who is shielded?

- Newborns
- Immunocompromised individuals
- Cancer patients on chemotherapy

Community protection depends on high coverage.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

07

Incomplete control points to access, supply, and trust barriers

Pneumococcal

40.5% case reduction

31.3% death reduction

Hepatitis A

89% case reduction

88.7% death reduction

Acute hepatitis B

83.9% case reduction

83.6% death reduction

System barriers named in the brief

Access gaps

Not everyone has access to vaccines globally

Supply shortages

Shortages put children at risk

Trust and uptake

Poor scientific understanding can reduce trust and uptake

Policy implication: closing the remaining gap requires improved access, coverage, trust, and newer vaccines.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

08

Priorities for resilient global vaccination systems

1

Elimination is achievable

100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.

2

Development has accelerated

Timelines collapsed from over 100 years for typhoid to under 1 year for COVID-19.

3

Mandates changed measles dynamics

Measles dropped to near zero across all 50 US states after the 1980 kindergarten mandate.

4

Eradication is rare but possible

Smallpox remains the only human disease eradicated through vaccination, certified in 1980.

5

Gaps still demand action

Pneumococcal disease, hepatitis A, and hepatitis B show persistent incomplete control.

Call to action

Improve access, coverage, and trust in vaccination — so global health systems can convert scientific progress into population protection.